

1(a)	1 irregular shape of cell (surface) membrane / cell loses its biconcave shape ; A cell swells / cell is larger / cell expands / AW I cell becomes flaccid / loses turgor / AW any two from: 2 cell membrane bursts / cell bursts / cell dies / cell lysis / haemolysis ; A short(er) life span 3 daughter cells, fill / AW, the cell ; 4 cell loses haemoglobin / haemoglobin used up (by pathogen) ; A pathogen uses amino acids from haemoglobin 5 less / no, oxygen transported ; A reduced / decreased, uptake of oxygen 6 release of toxins (by pathogen) ; I change in SA:V ratio / change of water potential	3
1(b)	any three from: 1 pathogen, feeds / grows / synthesises (named) biochemical(s) ; A increase in, volume / mass, of cytoplasm 2 production of (named) organelles ; 3 <u>DNA</u>, (semi-conservative) replication / synthesis ; R if given in incorrect phase 4 mitosis / mitoses / four named stages of mitosis ; 5 production of, several / many, nuclei ; 6 cytokinesis / described e.g. as <i>ref. to</i> formation of cleavage furrow ; I division of the cytoplasm 7 daughter cells 'bud off' (from the 'parent') ; A budding	3

Question	Answer	Marks
1(c)	any three from: 1 antibodies bind to, antigens / epitopes ; A attach / combine / join A form antibody-antigen complex I 'antibodies bind to pathogen' 2 complementary shape to antigens (on surface of malarial pathogen) ; 3 prevent, pathogen / infective stage, entering, red blood cells / liver cells ; 4 help / facilitate / AW, destruction by / mark pathogen for, phagocytes ; A stimulates phagocytosis / opsonisation in context / (leads to) lysis of pathogen 5 AVP ; e.g. phagocytes have receptors for constant region of antibodies e.g. <i>ref. to</i> agglutination / described I T-cells / production of antibodies by B / plasma cells I 'receptors on antibodies' / 'fight' and 'attack'	3

Question	Answer	Marks
1(d)	any five from: 1 access to, whole population / all people at risk ; 2 <i>idea that</i> reaching enough of the population to give herd immunity ; I herd immunity unqualified 3 vaccinating children, early enough in their lives / before the time when they are most at risk ; 4 <i>ref. to</i> education about, importance / benefits / advantages / AW, of vaccination ; 5 countering, anti-vaccination campaigns / misinformation about vaccines ; A <i>ref. to</i> 'anti-vaxxers' 6 long duration of protection given by, vaccine / artificial immunity ; A <i>ref. to</i> use of a 'live vaccine' 7 little / no, mutation of pathogen (to evade vaccine) ; A little / no, antigenic shift A few / no, strains develop I 'resistance to vaccine' 8 ability to change vaccine in response to changing strain(s) of pathogen ; 9 plentiful / good / AW, supply of vaccine ; 10 sufficient numbers of people trained to deliver vaccine ; A <i>idea that</i> administering vaccine is simple process 11 any positive reference to cost ; e.g. cost of, production / transport / storage / delivery to population A vaccine is provided free 12 long shelf life of vaccine ; 13 stable vaccine(s) ; e.g. in high temperatures A <i>ref. to</i> vaccine can be freeze-dried 14 no boosters required ; A easy to find people who need boosters 15 AVP ; e.g. governments / health authorities / countries, must have enough money to fund programme use of contact tracing to find people, at risk of disease / who should be offered vaccine / AW provide incentives for people to get vaccinated / <i>idea that</i> disease is a current threat to population no side effects	5